1 Assignment 1

1. What is the time for light to cross a proton?

   Solution:

   considering the diameter of proton $d = 1.6 \times 10^{-15}$ m

$$\begin{aligned} t &= \frac{d}{c} \\ &= \frac{1.6 \times 10^{-15}}{3 \times 10^{8}} \\ &= 5.33 \times 10^{-24}\text{s} \end{aligned}$$

2. Using the property of the metric tensor, show that:

$$A \cdot B = A^0 B^0 - \vec{A} \cdot \vec{B}$$

   Solution:

$$\begin{aligned} A.B &= g_{\mu\nu} A^{\mu} B^{\nu} \\ &= A^0 B^0 - \delta_{ij} A^i B^j \\ &= A^0 B^0 - \vec{A}.\vec{B} \end{aligned}$$

3. Show that:

$$\Lambda^{\alpha}{}_{\mu} \Lambda^{\beta}{}_{\nu} \eta_{\alpha\beta} = \eta_{\mu\nu}$$

   is a necessary condition for a Lorentz transformation.

   Solution:

   In Lorentz transformation $ds^2 = \eta_{\mu\nu} dx^{\mu} dx^{\nu}$ is invariant hence

$$\begin{aligned} \eta_{\mu\nu} dx^{\mu} dx^{\nu} &= \eta_{\mu\nu} dx'^{\mu} dx'^{\nu} \\ &= \eta_{\mu\nu} \Lambda^{\mu}_{\sigma} \Lambda^{\nu}_{\rho} dx^{\sigma} dx^{\rho} \end{aligned}$$

   and therefore

$$\boxed{\eta_{\mu\nu} = \eta_{\rho\sigma} \Lambda^{\rho}_{\mu} \Lambda^{\sigma}_{\nu}}$$

4. Relativistic wave equations contain derivatives with respect to space-time. How does a derivative with respect to space-time transform under a Lorentz transformation?

   Solution:

The covariant vector transforms according to

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$$

in Lorentz transformation
derivative with respect to spacetime ∂_{μ} is a covariant vector so they transform as

$$\partial'_{\mu} = \Lambda_{\mu}{}^{\nu} \partial_{\nu}$$

where $\Lambda_{\mu}{}^{\nu} = (\Lambda^{\mu}{}_{\nu})^{-1}$

5. A particle of mass M, initially at rest, decays into two pieces, each of mass m. What is the speed of each particle?
   Solution:
   Using momentum conservation we'll have momenta of both particles to be same
   Now energy conservation will give

$$Mc^2 = 2\gamma mc^2$$
$$\frac{1}{\sqrt{1 - v^2/c^2}} = \frac{M}{2m}$$
$$\frac{v^2}{c^2} = 1 - \frac{4m^2}{M^2}$$

$$\boxed{v = c\sqrt{1 - \frac{4m^2}{M^2}}}$$

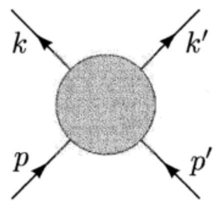
6. Consider the decay $X \to a + b$ where b is massless. What is the speed of a?
   Solution:
   Let mass of X be M and that of a be m. Assume particle X initially at rest, therefore both a and b will have same momentum $\mathbf{p}$

Using Energy Conservation

$$
\begin{aligned}
M &= \sqrt{p^2+m^2}+p \\
(M-p)^2 &= p^2+m^2 \\
M^2-2Mp &= m^2 \\
2M\gamma mv &= M^2-m^2 \\
\frac{v^2}{1-v^2} &= \left(\frac{M^2-m^2}{2Mm}\right)^2 \\
v &= \frac{\frac{M^2-m^2}{2Mm}}{\sqrt{1+\left(\frac{M^2-m^2}{2Mm}\right)^2}} \\
v &= \frac{M^2-m^2}{\sqrt{4M^2m^2+(M^2-m^2)^2}}
\end{aligned}
$$

$$\boxed{v = \frac{M^2-m^2}{M^2+m^2}}$$

7. Show that the Mandelstam variables s, t, u are invariant.



- $$
\begin{aligned}
s &= (p+p')^2 \\
&= (p+p')_\mu(p+p')^\mu \\
&= p^\mu p_\mu + p'^\mu p'_\mu + 2p^\mu p'_\mu \\
&= m_p^2 + m_{p'}^2 + 2p^\mu p'_\mu
\end{aligned}
$$

  All $m_p^2, m_{p'}^2$ and $p^\mu p'_\mu$ are Lorentz Scalars, hence are invariant. Thus s is Lorentz invariant

- $$
\begin{aligned}
t &= (k-p)^2 \\
&= m_k^2 + m_p^2 - 2k^\mu p_\mu
\end{aligned}
$$

  Hence Lorentz Invariant

- $$\begin{aligned} u &= (k' - p)^2 \\ &= m_{k'}^2 + m_p^2 - 2k'^{\mu} p_{\mu} \end{aligned}$$

  Again Lorentz Invariant

8. Show that in a $2 \to 2$ scattering process:

   $$s + t + u = \sum_i m_i^2$$

   Solution:

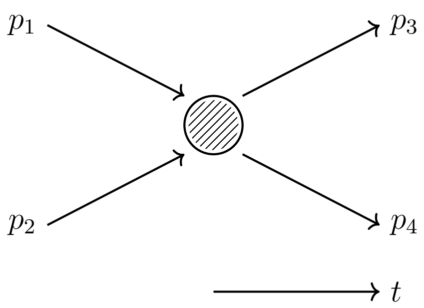


Figure 1:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ t &= (p_3 - p_1)^2 = (p_4 - p_2)^2 \\ u &= (p_4 - p_1)^2 = (p_3 - p_2)^2 \end{aligned}$$

$$\begin{aligned} s + t + u &= p_1^2 + p_2^2 + p_3^2 + p_1^2 + p_4^2 + p_1^2 + 2p_1.p_2 - 2p_1.p_3 - 2p_1.p_4 \\ &= 3p_1^2 + p_2^2 + p_3^2 + p_4^2 + 2p_1.(p_2 - p_3 - p_4) \end{aligned}$$

now by conservation of four momentum

$$\begin{aligned} p_1 + p_2 &= p_3 + p_4 \\ p_2 - p_3 - p_4 &= -p_1 \end{aligned}$$

therefore,

$$s + t + u = 3p_1^2 + p_2^2 + p_3^2 + p_4^2 - 2p_1^2$$
$$= p_1^2 + p_2^2 + p_3^2 + p_4^2$$

$$\boxed{s + t + u = \sum_i m_i^2}$$

9. Show that when all masses are neglected in a $2 \to 2$ reaction, then:

$$t = -\frac{s}{2}(1 - \cos\theta), \qquad u = -\frac{s}{2}(1 + \cos\theta)$$

where θ is the scattering angle.

Solution:

Scattering Angle is angle between incoming particle and outgoing particle in center of mass frame

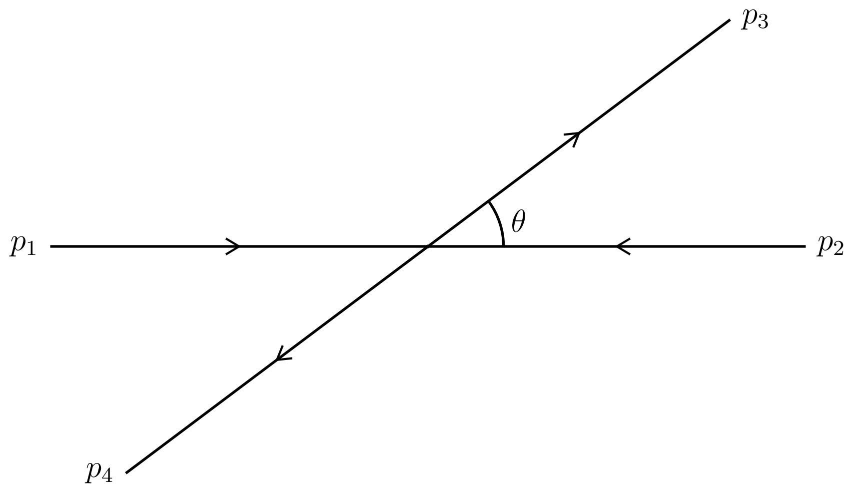


Figure 2: 2-2 Scattering in Center of Mass Frame

$$p_1 = (E, E, 0, 0, 0)$$
$$p_2 = (E, -E, 0, 0)$$
$$p_3 = (E, E\cos\theta, E\sin\theta, 0)$$
$$p_4 = (E, -E\cos\theta, -E\sin\theta, 0)$$

Now,

$$\begin{aligned} t &= (p_1 - p_3)^2 \\ &= -2p_1.p_2 \quad (\text{since } m_i = 0) \\ &= -2(E^2 - E^2 \cos\theta) \\ &= -2E^2(1 - \cos\theta) \end{aligned}$$

Similarly,

$$\begin{aligned} s &= (p_1 + p_2)^2 \\ &= 4E^2 \\ E^2 &= \frac{s}{4} \end{aligned}$$

Hence,

$$\boxed{t = -\frac{s}{4}(1 - \cos\theta)}$$

In similar way we can show

$$\begin{aligned} u &= (p_1 - p_4)^2 \\ &= -2p_1.p_4 \\ &= -2E^2(1 + \cos\theta) \end{aligned}$$

$$\boxed{u = -\frac{s}{4}(1 + \cos\theta)}$$

10. A particle of mass M, initially at rest, decays into two particles, each of mass m. What is the speed of each as they fly off?
Solution:
Same question as question 5

11. Show that in the rest frame of a decaying particle $a \to b + c$:

$$E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}$$

Solution:
we write four-momentum of particles as

$$p_a = (M_a, \mathbf{0}) \qquad p_b = (E_b, \mathbf{p}) \qquad p_c = (E_c, -\mathbf{p})$$

also,

$$E_b^2 = \mathbf{p}^2 + M_b^2 \tag{1}$$

$$M_a = E_b + E_c \tag{2}$$

Now

$$\begin{aligned}
p_a^2 &= p_b^2 + p_c^2 + 2p_a.p_b \\
M_a^2 &= M_b^2 + M_c^2 + 2(E_bE_c + \mathbf{p}^2) \\
&= M_b^2 + M_c^2 + 2(E_b(M_a - E_b) + E_b^2 - M_b^2) \qquad \text{(using (1) and (2))} \\
&= -M_b^2 + M_c^2 + 2M_aE_b
\end{aligned}$$

$$\boxed{E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}} \tag{3}$$

12. Suppose two identical particles, each of mass m and kinetic energy T, collide head-on. Find the kinetic energy T' in the fixed-target frame of reference.

Solution:

Let particles collide head on. To find T' consider the transformation of E in Lorentz transformation with boost such that one particle is at rest

$$E' = \gamma(E - p^1v^1)$$

We also have relations

$$p = \sqrt{T^2 + 2Tm}$$

$$v = \frac{p}{E} = \frac{\sqrt{T^2 + 2Tm}}{T + m}$$

Therefore,

$$\begin{aligned}
T' + m &= \frac{T + m + \sqrt{T^2 + 2Tm} \times \frac{\sqrt{T^2+2Tm}}{T+m}}{\sqrt{1 - \frac{T^2+2Tm}{(T+m)^2}}} \\
&= \frac{(T+m)^2 + T^2 + 2Tm}{m} \\
&= \frac{2T^2 + 4Tm}{m}
\end{aligned}$$

$$\boxed{T' = \frac{2T(T + 2m)}{m}}$$

13. Consider the decay $\pi^+ \to \mu^+\nu_\mu$, where $E_{\pi^+} = 20$ GeV and $\tau_{\pi^+} = 2.6 \times 10^{-8}$ s. What is the time dilation?

Solution:

We have

$$\gamma = \frac{E}{m_\mu}$$

Mass of Muon $m_\mu \approx 106$ MeV

$$\gamma = \frac{20000}{106} = 188.68$$

Therefore the time dilated

$$t' = 2.6 \times 10^{-8} \times 188.68 = 4.9$$

$$\boxed{t' = 4.9\ \mu\text{s}}$$

Time is dilated by factor of $\sim$188

14. In the scattering process $a+b \to c+d$, where θ^* is the scattering angle in the CMS, show that:

$$\cos\theta^* = \frac{s(t-u) + (m_a^2 - m_b^2)(m_c^2 - m_d^2)}{\lambda^{1/2}(s, m_a^2, m_b^2)\,\lambda^{1/2}(s, m_c^2, m_d^2)}$$

And for elastic scattering, show that:

$$t = -4|\vec{p}|^2 \sin^2\left(\frac{\theta^*}{2}\right)$$

where $\vec{p}$ is the initial momentum in the CMS

Solution:

We'll use the relations

$$E_a = \frac{s + m_a^2 - m_b^2}{2\sqrt{s}} \qquad E_b = \frac{s + m_b^2 - m_a^2}{2\sqrt{s}}$$

$$|\mathbf{p}| = \frac{\lambda^{1/2}(s, m_a^2, m_b^2)}{2\sqrt{s}} \qquad |\mathbf{p}'| = \frac{\lambda^{1/2}(s, m_c^2, m_d^2)}{2\sqrt{s}}$$

$$\begin{aligned}
t &= (p_a - p_c)^2 \\
&= m_a^2 + m_c^2 - 2(E_a E_c - |\mathbf{p}||\mathbf{p}'|\cos\theta) \\
u &= (p_a - p_d)^2 \\
&= m_a^2 + m_d^2 - 2(E_a E_d + |\mathbf{p}||\mathbf{p}'|\cos\theta) \\
t - u &= m_c^2 - m_d^2 - 2E_a(E_c - E_d) + 4|\mathbf{p}||\mathbf{p}'|\cos\theta
\end{aligned}$$

$E_c - E_d = \frac{m_c^2 - m_d^2}{\sqrt{s}}$

$$t - u = m_c^2 - m_d^2 - 2\left(\frac{s + m_a^2 - m_b^2}{2\sqrt{s}}\right)\left(\frac{m_c^2 - m_d^2}{\sqrt{s}}\right) + 4\frac{\lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)}{4s}\cos\theta$$
$$s(t-u) = s(m_c^2 - m_d^2) - \left(s + m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$
$$s(t-u) = -\left(s - s + m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$
$$s(t-u) = -\left(m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$

$$\boxed{\cos\theta = \frac{s(t-u) + \left(m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right)}{\lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)}}$$

For Elastic Scattering $a = c$ and $b = d$, therefore, $|\mathbf{p}| = |\mathbf{p}'|$

$$\begin{aligned} t &= (p_a - p_c)^2 \\ &= (E_a - E_c)^2 - (\mathbf{p_a} - \mathbf{p_c})^2 \\ &= -2(|\mathbf{p}|^2 - |\mathbf{p}|^2\cos\theta) \end{aligned}$$

$$\boxed{t = -4|\mathbf{p}|^2\sin^2\left(\frac{\theta}{2}\right)}$$

15. The decay width of the η-meson is $\Gamma_\eta = 2.6$ keV. What is the lifetime τ in seconds?
Solution:

$$\begin{aligned} \tau &= \frac{1}{\Gamma} \\ &= \frac{1}{2.6 \text{ keV}} \\ &= 0.3846 \ (\text{keV})^{-1} \end{aligned}$$

$$\begin{aligned} 1 \text{ s} &= 1.52 \times 10^{24} \ (\text{GeV})^{-1} \\ &= 1.52 \times 10^{18} \ (\text{keV})^{-1} \\ (\text{keV})^{-1} &= 6.58 \times 10^{-19} \text{ s} \\ \tau &= 0.3846 \times 6.58 \times 10^{-19} \end{aligned}$$

$$\boxed{\tau = 2.53 \times 10^{-19} \text{ s}}$$

16. The lifetime of π^0 is $\tau_{\pi^0} = 0.89 \times 10^{-16}$ s. What is the decay width Γ in natural units?
Solution:

$$0.89 \times 10^{-16}\ \text{s} = 0.1353\ (\text{eV})^{-1}$$

Decay Width

$$\Gamma = \frac{1}{\tau}$$
$$= \frac{1}{0.1353}$$

$$\boxed{\Gamma = 7.4\ \text{eV}}$$

17. Show that $d^4p = dp^0\, dp^1\, dp^2\, dp^3$ is a Lorentz-invariant quantity.
Solution:
d^4p will transform like a tensor-density in Lorentz transformation

$$d^4p' = \left|\left|\frac{\partial x'}{\partial x}\right|\right| d^4p$$

where $\left|\frac{\partial x'}{\partial x}\right|$ is the Jacobian of transformation, which is

$$\det(\Lambda^{\mu}{}_{\nu})$$

Since in a proper orthochronous Lorentz transformation det(Λ)=1, the quantity d^4p is Lorentz invariant

18. Consider the hyperon decay $\Lambda \to p + \pi^-$. Given:

$$M_\Lambda = 1.1157\ \text{GeV}, \quad M_p = 0.9383\ \text{GeV}, \quad m_{\pi^-} = 0.1396\ \text{GeV},$$

calculate the energy of the outgoing pion.
Solution:
We can use eqn. (3) from problem 11

$$E_\pi = \frac{M_\Lambda^2 - M_p^2 + M_\pi^2}{2M_\Lambda}$$

Therefore,

$$E_\pi = \frac{1.1157 - 0.9383 + 0.1396}{2 \times 1.1157}$$

$$\boxed{E_\pi = 0.142\ \text{GeV}}$$

19. Two particles of mass m each and speed v collide at right angles forming a new body of mass M. What is the new mass M?

Solution:

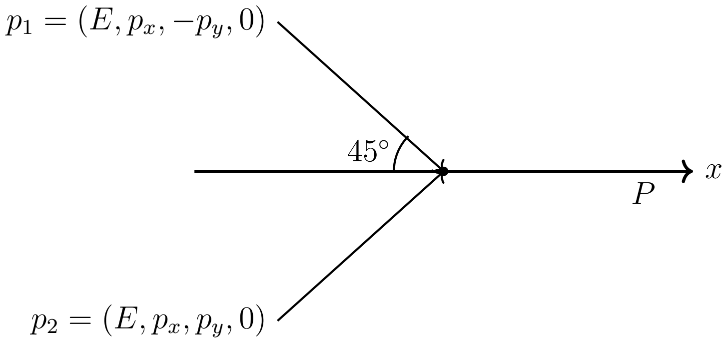


Figure 3:

Use $p^\mu p_\mu = m^2$ we have

$$
\begin{aligned}
p_1^2 + p_2^2 + 2p_1.p_2 &= M^2 \\
2m^2 + 2(E^2 - p_x^2 + p_y^2) &= M^2 \\
2m^2 + 2E^2 &= M^2 \qquad \left(\text{since } p_x = p_y = \frac{p}{\sqrt{2}}\right) \\
\frac{2m^2}{1-v^2} + 2m^2 &= M^2
\end{aligned}
$$

$$\boxed{M = m\sqrt{2\left(1 - \frac{1}{1-v^2}\right)}}$$

20. Show that rapidity η is additive.

Solution:

Relativistically the velocity in same direction is added according to

$$v = \frac{v + v'}{1 + \frac{vv'}{c^2}} \tag{4}$$

Now we define rapidity ξ as

$$\beta = \frac{v}{c} = \tanh\xi$$

we can now write

$$v = c\tanh\xi$$

$$v' = c\tanh\xi'$$

Equation (4) can be written as

$$
\begin{aligned}
&= \frac{c\tanh\xi + c\tanh\xi'}{1+\tanh\xi\tanh\xi'} \\
&= c\tanh(\xi+\xi')
\end{aligned}
$$

which shows that

$$\boxed{\xi_{\mathrm{res}} = \xi + \xi'}$$

and hence rapidity is additive

Particle Physics

Assignment – 2

1. Show that for any two vectors $\vec{A}$ and $\vec{B}$,

$$(\boldsymbol{\sigma} \cdot \vec{A})(\boldsymbol{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B}\, \mathbf{1} + i\boldsymbol{\sigma} \cdot (\vec{A} \times \vec{B}),$$

where $\boldsymbol{\sigma}$ are the Pauli matrices.

Solution

We'll use the relations

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \qquad \{\sigma_i, \sigma_j\} = 2\delta_{ij}$$

now

$$\begin{aligned} \sigma_i A_i \sigma_j B_j &= \left(\frac{1}{2}\{\sigma_i, \sigma_j\} + \frac{1}{2}[\sigma_i, \sigma_j]\right) A_i B_j \\ &= (\delta_{ij} + i\epsilon_{ijk}\sigma_k) A_i B_j \\ &= A_i B_i + i\sigma_k (\vec{A} \times \vec{B})_k \end{aligned}$$

and hence

$$\boxed{(\boldsymbol{\sigma} \cdot \vec{A})(\boldsymbol{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B}\, \mathbf{1} + i\boldsymbol{\sigma} \cdot (\vec{A} \times \vec{B})}$$

2. Show that the ratio of the decay widths of

$$\frac{\Gamma(\Delta^{++} \to p\pi^+)}{\Gamma(\Delta^+ \to n\pi^+)} = 3,$$

using the Clebsch–Gordon coefficients.

Solution:

We will consider $|I\ I_3\rangle$ basis for particles.

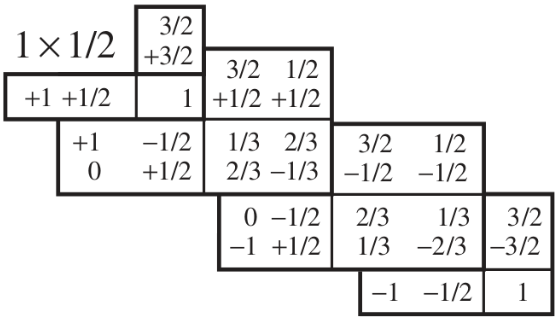
We will consider the reaction rate is

$$\sigma \sim |\text{amplitude}|^2 \sim \sum_I |\langle I'\ I_3'|\, A\, |I\ I_3\rangle|^2$$

The matrix elements can be calculated using Wigner-Eckart Theorem and will be proportional to Clebsh-Gordon coefficients.

We have $|\Delta^{++}\rangle = \left|\frac{3}{2}\frac{3}{2}\right\rangle$ and $|\Delta^{+}\rangle = \left|\frac{3}{2}\frac{1}{2}\right\rangle$ since p has $I_3 = 1/2$ and π^+ 1 we get $|p\pi^+\rangle = \left|\frac{3}{2}\frac{3}{2}\right\rangle$.

now since $|\pi^+ n\rangle = \left|1 - \frac{1}{2}\right\rangle$ in $|I_1\ I_2\rangle$ basis we need to find it's representation in $|I\ I_3\rangle$



$$\left|\frac{3}{2}\ \frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}\left|1 - \frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|0\ \frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2}\ \frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|1 - \frac{1}{2}\right\rangle - \frac{1}{\sqrt{3}}\left|0\ \frac{1}{2}\right\rangle$$

and therefore

$$\left|1 - \frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|\frac{1}{2}\ \frac{1}{2}\right\rangle + \frac{1}{\sqrt{3}}\left|\frac{3}{2}\ \frac{1}{2}\right\rangle = |\pi^+\ n\rangle$$

now

$$\frac{\Gamma(\Delta^{++} \to p\pi^+)}{\Gamma(\Delta^+ \to n\pi^+)} = \left|\frac{\left\langle\frac{3}{2}\frac{3}{2}\middle|\frac{3}{2}\frac{3}{2}\right\rangle}{\left\langle\frac{3}{2}\frac{1}{2}\right|\left(\sqrt{\frac{2}{3}}\left|\frac{1}{2}\ \frac{1}{2}\right\rangle + \frac{1}{\sqrt{3}}\left|\frac{3}{2}\ \frac{1}{2}\right\rangle\right)}\right|^2$$

$$= \left(\frac{1}{\frac{1}{\sqrt{3}}}\right)^2$$

and therefore

$$\boxed{\frac{\Gamma(\Delta^{++} \to p\pi^+)}{\Gamma(\Delta^+ \to n\pi^+)} = 3}$$

3. Suppose the pion–nucleon interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = g\bar{N}\gamma_5\tau^a N\pi^a,$$

where g is the coupling constant, τ^a are the Pauli matrices, N is the nucleon field, and $\pi^a = (\pi^1, \pi^2, \pi^3)$. Identify the charged and neutral pions in terms of π^1, π^2, π^3.

4. Show that the structure constants f^{abc} of the SU(3) Lie algebra can be written as

$$f^{abc} = \frac{1}{4i}\,\text{Tr}([\lambda^a, \lambda^b]\lambda^c).$$

Solution:

$$
\begin{aligned}
\frac{1}{4i}\mathrm{Tr}([\lambda^a,\lambda^b]\lambda^c) &= \frac{1}{2}f^{abd}\ \mathrm{Tr}(\lambda^d\lambda^c) \\
&= f^{abd}\delta_{dc} \\
&= f^{abc}
\end{aligned}
$$

where we have used the fact that $\mathrm{Tr}(\lambda^a\lambda^c) = 2\delta_{ab}$ and $[\lambda^a,\lambda^b] = 2if^{abc}\lambda^c$

5. Show that the Gell–Mann matrices λ^a satisfy

$$\{\lambda^a,\lambda^b\} = C^{ab}\mathbf{1} + 2d^{abc}\lambda^c,$$

where

$$C^{ab} = \frac{4}{3}\delta^{ab},$$

and

$$d^{abc} = \frac{1}{4}\ \mathrm{Tr}(\{\lambda^a,\lambda^b\}\lambda^c).$$

Solution:

We'll again use the property $\mathrm{Tr}(\lambda^a\lambda^c) = 2\delta_{ab}$ We can write any arbitrary matrix as

$$
\begin{aligned}
M &= a_0\mathbb{1} + a_\alpha\lambda^\alpha \\
&\text{(Put M as } \{\lambda^a,\lambda^b\}) \\
\mathrm{Tr}(\{\lambda^a,\lambda^b\}) &= 3a_0 \\
a_0 &= \frac{1}{3}\ \mathrm{Tr}(\lambda^a\lambda^b + \lambda^b\lambda^a) \\
a_0 &= \frac{2}{3}\ \mathrm{Tr}(\lambda^a\lambda^b) \\
a_0 &= \frac{4}{3}\delta_{ab}
\end{aligned}
$$

Now multiply first equation with λ^c and then take trace

$$
\begin{aligned}
\mathrm{Tr}(\{\lambda^a,\lambda^b\}\lambda^c) &= a_0\mathrm{Tr}(\lambda^c) + a_\alpha\mathrm{Tr}(\lambda^\alpha\lambda^c) \\
\mathrm{Tr}(\{\lambda^a,\lambda^b\}\lambda^c) &= 2a_\alpha\delta_{\alpha c} \\
a_c &= \frac{1}{2}\mathrm{Tr}(\{\lambda^a,\lambda^b\}\lambda^c)
\end{aligned}
$$

And hence

$$\boxed{\{\lambda^a,\lambda^b\} = \frac{4}{3}\delta_{ab}\mathbb{1} + \frac{1}{2}\mathrm{Tr}(\{\lambda^a,\lambda^b\}\lambda^c)\lambda^c}$$

6. Using Young tableaux, find the irreducible representations for:

(a) $6 \otimes 3$ of SU(3),

Solution:

6 corresponds to $\ydiagram{2}$ therefore,

$$\ydiagram{2} \otimes \ydiagram{1} = \begin{ytableau} 3 & 4 & 5 \end{ytableau} \oplus \begin{ytableau} 3 & 4 \\ 2 \end{ytableau}$$

where the dimension of first diagram will be

$$d = \frac{3 \times 4 \times 5}{3 \times 2} = 10$$

and of second diagram

$$d = \frac{4 \times 3 \times 2}{3} = 8$$

therefore

$$\boxed{6 \otimes 3 = 10 \oplus 8}$$

(b) $4 \otimes 4 \otimes 4$ of SU(4),

Solution:

$$\ydiagram{1} \otimes \ydiagram{1} = \ydiagram{2} \oplus \ydiagram{1,1}$$

Therefore

$$\begin{aligned}
&\left(\ydiagram{2} \oplus \ydiagram{1,1}\right) \otimes \ydiagram{1} \\
&= \ydiagram{3} \oplus \ydiagram{2,1} \oplus \ydiagram{2,1} \oplus \ydiagram{1,1,1} \\
&= \begin{ytableau} 4 & 5 & 6 \end{ytableau} \oplus \begin{ytableau} 4 & 5 \\ 3 \end{ytableau} \oplus \begin{ytableau} 4 & 5 \\ 3 \end{ytableau} \oplus \begin{ytableau} 4 \\ 3 \\ 2 \end{ytableau}
\end{aligned}$$

The dimension of first diagram is

$$\frac{4 \times 5 \times 6}{2 \times 3} = 20$$

second and third is

$$\frac{4 \times 5 \times 3}{3} = 20$$

and last one is 4 therefore

$$\boxed{4 \otimes 4 \otimes 4 = 20 \oplus 20 \oplus 20 \oplus \bar{4}}$$

(c) $4 \otimes \bar{4}$ of SU(4).

Solution:

$$\square \otimes \begin{array}{|c|}\hline \\ \hline \\ \hline \\ \hline \end{array} = \begin{array}{|c|}\hline 1 \\ \hline \end{array} \oplus \begin{array}{|c|c|}\hline 4 & 5 \\ \hline 3 \\ \cline{1-1} 2 \\ \cline{1-1} \end{array}$$

with

$$d_1 = 1$$

and

$$d_2 = \frac{4 \times 5 \times 3 \times 2}{4 \times 2} = 15$$

therefore

$$\boxed{4 \otimes \bar{4} = 1 \oplus 15}$$

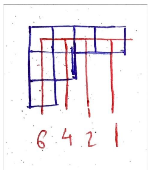
7. Determine the dimension of the irreducible representations shown by the Young tableaux:

(a) $\begin{array}{|c|c|c|c|}\hline & & & \\ \hline & \\ \cline{1-2} \\ \cline{1-1} \end{array}$ of SU(3),

Solution:

$$\begin{array}{|c|c|c|c|}\hline 3 & 4 & 5 & 6 \\ \hline 2 & 3 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array}$$

multiplying all these numbers we get numerator while finding dimensions to get the denominator we use hook technique



therefore

$$d = \frac{3 \times 4 \times 5 \times 6 \times 2 \times 3}{6 \times 4 \times 2}$$

$$\boxed{d = 30}$$

(b) $\begin{array}{|c|c|c|}\hline & & \\ \hline \end{array}$ of SU(4).

Solution:

$$\begin{array}{|c|c|c|}\hline 4 & 5 & 6 \\ \hline \end{array}$$

We get

$$d = \frac{4 \times 5 \times 6}{2 \times 3}$$

$$\boxed{d = 20}$$

8. Determine the Young tableaux for the direct products:

   (a) $\ydiagram{2,1} \otimes \ydiagram{1,1}$ of SU(3),
   Solution:

$$\ydiagram{2,1} \otimes \ytableaushort{a,b} =$$

$$\left(\ytableaushort{{}{}a,{}} \oplus \ytableaushort{{}{},{}a} \oplus \ytableaushort{{}{},{},a} \right) \otimes \ytableaushort{b}$$

$$= \ytableaushort{{}{}a,{},b} \oplus \ytableaushort{{}{}a,{}b} \oplus \ytableaushort{{}{}ab,{}} \oplus$$

$$\ytableaushort{{}{},{}a,b} \oplus \ytableaushort{{}{}b,{}a} \oplus$$

$$\ytableaushort{{}{}b,{},a} \oplus \ytableaushort{{}{},{}b,a}$$

Now using the rule that: *Reading along the rows from right to left from the top row down to the bottom row, the number of a's must be greater than or equal to the number of b's*, We get final anwer

$$\boxed{\ydiagram{2,1} \otimes \ydiagram{1,1} = \ydiagram{2} \oplus \ydiagram{3,2} \oplus \ydiagram{1,1}}$$

i.e

$$8 \otimes 3 = 6 \oplus 15 \oplus \bar{3}$$

   (b) $\ydiagram{2,1} \otimes \ydiagram{2}$ of SU(3).
   Solution:

$$\ydiagram{2,1} \otimes \ytableaushort{aa} =$$

$$\left(\ytableaushort{{}{}a,{}} \oplus \ytableaushort{{}{},{}a} \oplus \ytableaushort{{}{},{},a} \right) \otimes \ytableaushort{a}$$

$$= \ytableaushort{{}{}a,{},a} \oplus \ytableaushort{{}{}aa,{}} \oplus \ytableaushort{{}{}a,{}a} \oplus$$

$$\ytableaushort{{}{},{}a,a} \oplus \ytableaushort{{}{}a,{}a} \oplus$$

$$\ytableaushort{{}{}a,{},a} \oplus \ytableaushort{{}{},{}a,a}$$

$$= \ytableaushort{{}a} \oplus \ytableaushort{{}{}aa,{}} \oplus \ytableaushort{{}{}a,{}a} \oplus \ytableaushort{{},a} \oplus$$

$$\ytableaushort{{}a{},\none\none a}$$

Now removing similar diagrams we get final answer

$$\boxed{\ydiagram{2,1} \otimes \ydiagram{2} = \ydiagram{3,2} \oplus \ydiagram{1,1} \oplus \ydiagram{4,1} \oplus \ydiagram{2}}$$

i.e.

$$8 \otimes 6 = 15 \oplus \bar{3} \oplus 24 \oplus 6$$

9. If $\ydiagram{2}$ represents the 6 of SU(3), what does $\ydiagram{2,2}$ represent?

Solution:

if we add the first diagram just below the second one we get

$$\ytableaushort{{}{},{}{},aa} = 1$$

i.e. singlet. hence $\ydiagram{2,2}$ represent the conjugate representation 6^*

10. Draw the weight diagrams for SU(3):

Name all the particles and their quark compositions if $3 = (u, d, s)$ and determine the corresponding quantum numbers.

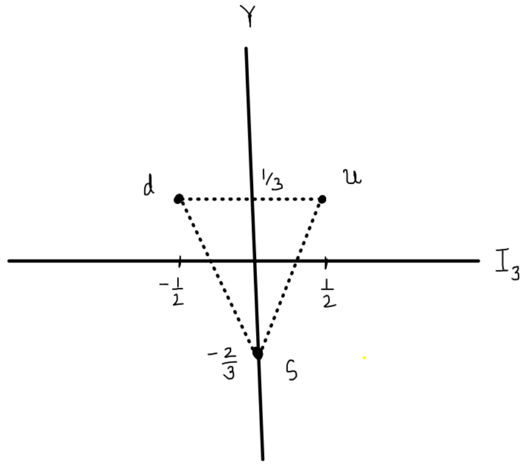
(a) $3 \otimes 3 \otimes 3$

Solution:

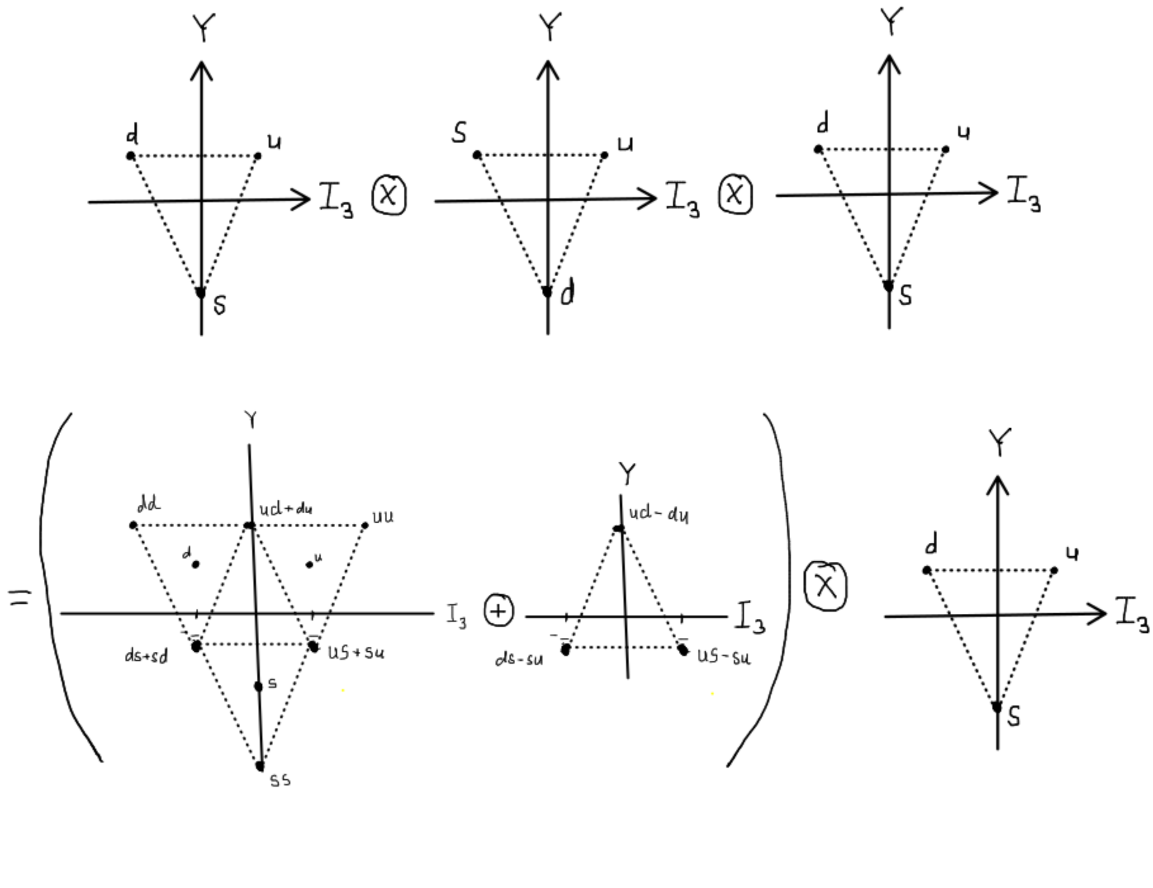
We have two diagonal generators in SU(3):

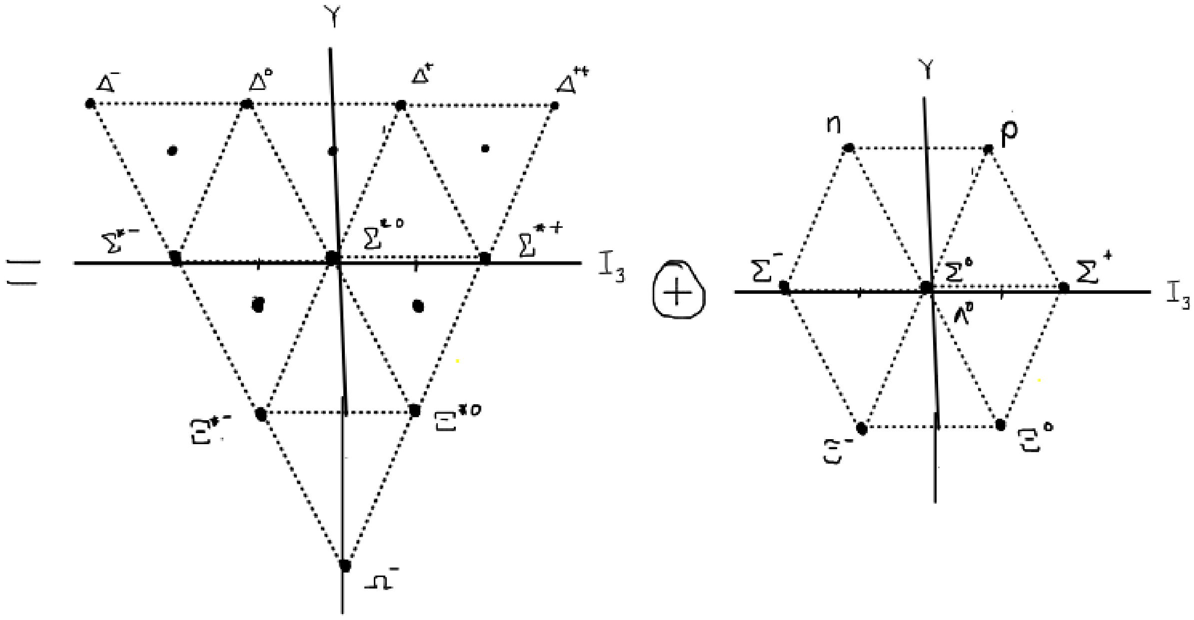
$$\lambda_3 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \lambda_8 = \frac{1}{2\sqrt{3}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

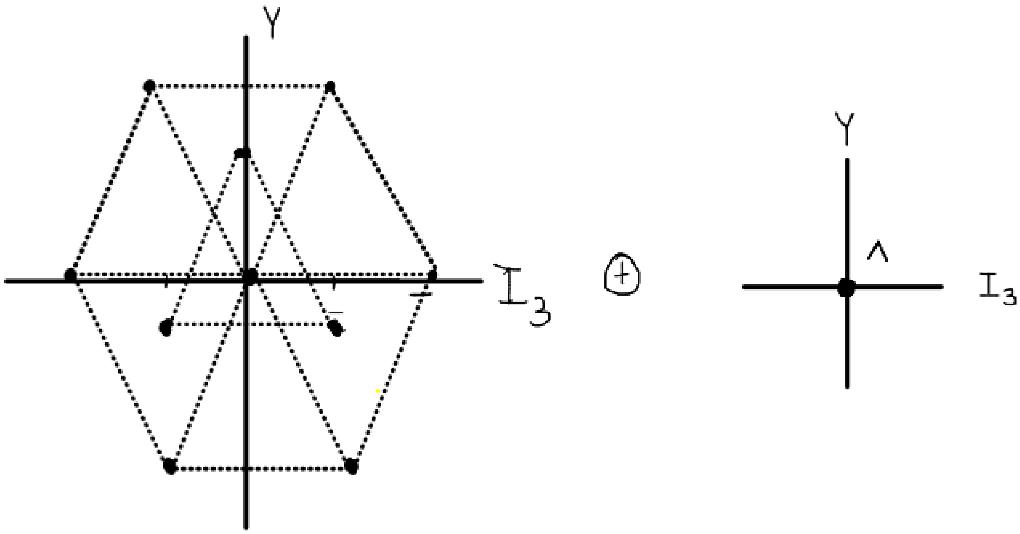
Hence we have 3 root vectors $(\frac{1}{2}, \frac{1}{2\sqrt{3}}), (-\frac{1}{2}, \frac{1}{2\sqrt{3}})$ and $(0, -\frac{2}{2\sqrt{3}})$. From which we can weight diagram



now for $3 \otimes 3 \otimes 3$



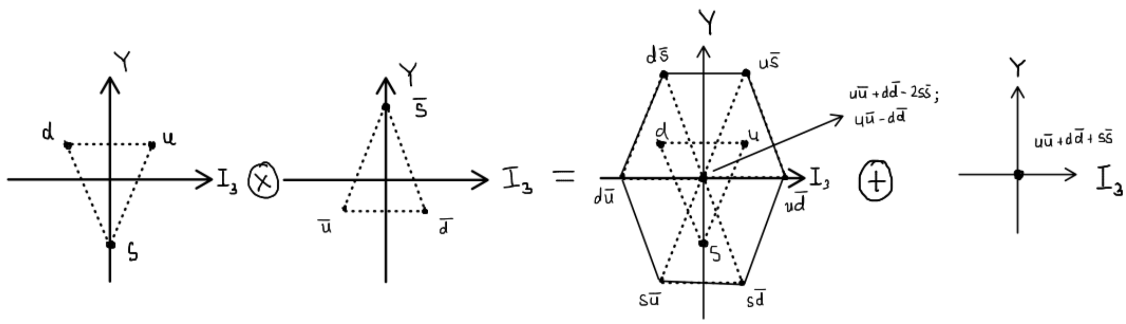




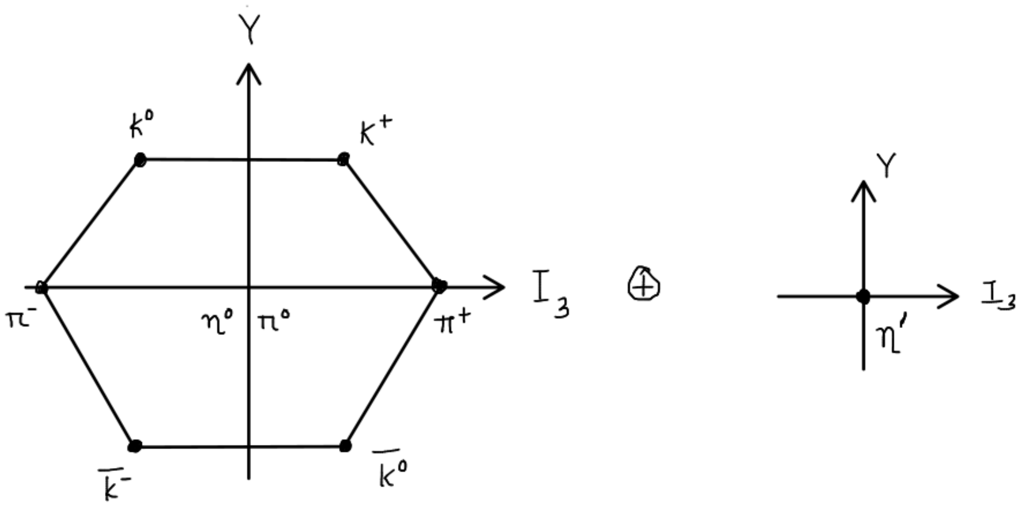
(b) $3 \otimes \bar{3}$.

Solution:

We know $3 \otimes \bar{3} = 8 \oplus 1$. Therefore weight diagrams will look like



With particles being



11. For two particles with isospin $I = 2$ and $I = \frac{1}{2}$, write down the states using Clebsch–Gordon decomposition.

(a) $|2,2\rangle \left|\frac{1}{2},-\frac{1}{2}\right\rangle$ in terms of $\left|\frac{5}{2},\frac{3}{2}\right\rangle$ and $\left|\frac{3}{2},\frac{3}{2}\right\rangle$ states,

Solution:

$$\left|\frac{5}{2},\frac{5}{2}\right\rangle = |2,2\rangle \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\text{(Operate by } J_- = J_{1-} + J_{2-}\text{)}$$

$$\left|\frac{5}{2},\frac{3}{2}\right\rangle = \frac{2}{\sqrt{5}}|2,1\rangle \left|\frac{1}{2},\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}|2,2\rangle \left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{5}$$

The state $\left|\frac{3}{2},\frac{3}{2}\right\rangle$ will be orthogonal to $\left|\frac{5}{2},\frac{3}{2}\right\rangle$ and hence can be given by

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \frac{1}{\sqrt{5}}|2,1\rangle \left|\frac{1}{2},\frac{1}{2}\right\rangle - \frac{2}{\sqrt{5}}|2,2\rangle \left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{6}$$

We can invert the two to obtain the desired state giving

$$\boxed{|2,2\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle=\frac{1}{\sqrt{5}}\left|\frac{5}{2},\frac{3}{2}\right\rangle-\frac{2}{\sqrt{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle}$$

(b) $|2,-1\rangle\left|\frac{1}{2},\frac{1}{2}\right\rangle$ in terms of $\left|\frac{5}{2},-\frac{1}{2}\right\rangle$ and $\left|\frac{3}{2},-\frac{1}{2}\right\rangle$ states,
Solution:
Applying J_- twice on (5) we get

$$\left|\frac{5}{2},\frac{1}{2}\right\rangle=\sqrt{\frac{2}{5}}|2,-1\rangle\left|\frac{1}{2},\frac{1}{2}\right\rangle+\sqrt{\frac{3}{5}}|2,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{7}$$

Applying J_- once on the state (6) we get

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle=\sqrt{\frac{3}{5}}|2,-1\rangle\left|\frac{1}{2},\frac{1}{2}\right\rangle-\sqrt{\frac{2}{5}}|2,0\rangle\left|\frac{1}{2},-\frac{1}{2}\right\rangle \tag{8}$$

Now inverting these two we obtain

$$\boxed{|2,-1\rangle\left|\frac{1}{2},\frac{1}{2}\right\rangle=\sqrt{\frac{2}{5}}\left|\frac{5}{2},-\frac{1}{2}\right\rangle+\sqrt{\frac{3}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle}$$

12. The SU(3) generators satisfy

$$[F^a,F^b]=if^{abc}F^c.$$

We identify two diagonal generators F^3 and F^8 satisfying $[F^3,F^8]=0$. Define

$$T^{\pm}=F^1\pm iF^2,\quad U^{\pm}=F^6\pm iF^7,\quad V^{\pm}=F^4\pm iF^5,$$

with

$$T^3=F^3,\quad Y=\frac{2}{\sqrt{3}}F^8.$$

They satisfy the commutation relations:

$$[T^3,T^{\pm}]=\pm T^{\pm},\qquad [Y,T^{\pm}]=0,$$

$$[T^3,U^{\pm}]=\mp\frac{1}{2}U^{\pm},\qquad [Y,U^{\pm}]=\pm U^{\pm},$$

$$[T^3,V^{\pm}]=\pm\frac{1}{2}V^{\pm},\qquad [Y,V^{\pm}]=\pm V^{\pm},$$

$$[T^+, T^-] = 2T^3,$$

$$[U^+, U^-] = \frac{3}{2}Y - T^3,$$

$$[V^+, V^-] = \frac{3}{2}Y + T^3,$$

$$[T^+, V^-] = -U^-, \quad [T^-, U^+] = V^+, \quad [U^+, V^-] = T^-.$$

Determine all the non-vanishing structure constants f^{abc} using the above commutation relations.

Solution:

Along with the commutation we will use property that f^{abc} is completely antisymmetric for SU(3)

Proof:

by definition $f^{abc} = f^{acb}$

now

$$\begin{aligned}\mathrm{Tr}([T^a, T^b]T^c) &= \mathrm{Tr}(if^{abd}T^dT^c) \\ &= \frac{i}{2}f^{abd}\delta^{dc} \\ &= \frac{i}{2}f^{abc}\end{aligned}$$

$$\begin{aligned}\mathrm{Tr}([T^a, T^b]T^c) &= -\mathrm{Tr}([T^b, T^a]T^c]) \\ &= -\frac{i}{2}f^{bac}\end{aligned}$$

Therefore $f^{abc} = -f^{bac}$

by using cyclic property of trace

$$\begin{aligned}\mathrm{Tr}([T^a, T^b]T^c) &= \mathrm{Tr}(T^a[T^b, T^c]) \\ &= -\mathrm{Tr}(T^a[T^c, T^b]) \\ &= -\frac{i}{2}f^{cba}\end{aligned}$$

Therefore $f^{abc} = -f^{cba}$

Therefore *Structure constants for SU(3) are completely antisymmetric*

$$[F^3, F^1 + iF^2] = if^{31a}F^a - f^{32a}F^a = F^1 + iF^2$$

Comparing we get

$$f^{12\alpha} = \begin{cases} 1 & \text{for } \alpha = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$[F^3, F^6 + F^7] = -\frac{1}{2}(F^6 + iF^7)$$

Therefore

$$f^{37\alpha} = \begin{cases} 1/2 & \text{for } \alpha = 6 \\ 0 & \text{otherwise} \end{cases}$$

$$[F^3, F^4 + iF^5] = if^{34a}F^a - f^{35a}F^a = \frac{1}{2}(F^4 + iF^5)$$

Therefore,

$$f^{34\alpha} = \begin{cases} 1/2 & \text{for } \alpha = 5 \\ 0 & \text{otherwise} \end{cases}$$

$$[2\sqrt{3}F^8, F^1 + iF^2] = 2\sqrt{3}(if^{81a}F^a - f^{82a}F^a) = 0$$

Therefore $f^{81a} = f^{82a} = 0 \quad \forall\ a$

$$\left[\frac{2}{\sqrt{3}}F^8, F^6 + iF^7\right] = F^6 + iF^7$$
$$if^{86a}F^a - f^{82a}F^a = F^6 + iF^7$$

$$f^{86\alpha} = \begin{cases} \sqrt{3}/2 & \text{for } \alpha = 7 \\ 0 & \text{otherwise} \end{cases}$$

$$\left[\frac{2}{\sqrt{3}}F^8, F^4 + iF^5\right] = F^4 + iF^5$$
$$if^{84a}F^a - f^{85a}F^a = F^4 + iF^5$$

$$f^{84\alpha} = \begin{cases} \sqrt{3}/2 & \text{for } \alpha = 5 \\ 0 & \text{otherwise} \end{cases}$$

Use $[T_+, V_-] = -U_-$

$$\left[F^1 + iF^2, F^4 - iF^5\right] = if^{14a}F^a + f^{15a}F^a - f^{24a}F^a + if^{25a}F^a = -F^6 + iF^7$$

$$f^{147} + f^{257} = 1 \tag{9}$$

$$f^{156} - f^{246} = -1 \tag{10}$$

This tells us that

$$f^{14\alpha} = 0 \quad \alpha \neq 7$$

$$f^{25\alpha} = 0 \quad \alpha \neq 7$$

$$f^{15\alpha} = 0 \quad \alpha \neq 6$$

$$f^{24\alpha} = 0 \quad \alpha \neq 6$$

Use $[T_-, U_+] = V_+$.

$$\left[F^1 - iF^2, F^6 + iF^7\right] = -F^4 - iF^5$$

which gives

$$f^{165} + f^{275} = -1 \tag{11}$$

$$f^{264} - f^{174} = -1 \tag{12}$$

$$\left[F^6 + iF^7, F^4 - iF^5\right] = F^1 - iF^2$$

This gives

$$f^{642} + f^{752} = -1 \tag{13}$$

$$f^{651} - f^{741} = 1 \tag{14}$$

solving (5),(6),(9) and (10) we get $f^{147} = f^{165} = f^{245} = f^{257} = 1/2$ Therefore the non-vanishing Structure constants are

$$\boxed{f_{123} = 1 \qquad f_{458} = f_{678} = \sqrt{3}/2}$$

$$\boxed{f_{147} = f_{165} = f_{246} = f_{257} = f_{345} = f_{376} = 1/2}$$

Particle Physics

Assignment – 3

Problem 1

Consider a process $a + b \to c + d$ where particle b is the target at rest in the lab. Let E_a^L be the lab energy of the beam particle and E_a^* be the CoM energy of a.

(a) Derive the Lorentz boost that connects the lab and CoM frames. Show that the boost velocity and Lorentz factor are:

$$\beta_{cm} = \frac{|\mathbf{p}_a^L|}{E_a^L + m_b}, \qquad \gamma_{cm} = \frac{E_a^L + m_b}{\sqrt{s}}.$$

Solution:
We have four momentum for the particles in the lab frames as
$p_a^\mu = (E_a^L, \mathbf{p}_a^L)$ and $p_b^\mu = (m_b, \mathbf{0})$ $\qquad P^\mu = p_a^\mu + p_b^\mu = (E_a^L + m_b, \mathbf{p}_a^L)$
In the COM frame the whole system will have net momentum $P' = 0$. Using transformation of spacial momentum under Lorentz transformation

$$\begin{aligned} P' &= \gamma(P - \beta E) \\ 0 &= |\mathbf{p}_a^L| - \beta_{cm}(E_a^L + m_b) \end{aligned}$$

$$\boxed{\beta_{cm} = \frac{|\mathbf{p}_a^L|}{E_a^L + m_b}} \tag{15}$$

$$\begin{aligned} \gamma_{cm} &= \frac{1}{\sqrt{1-\beta^2}} \\ &= \frac{1}{\sqrt{1 - \frac{|\mathbf{p}_a^L|^2}{(E_a^L + m_b)^2}}} \\ &= \frac{E_a^L + m_b}{\sqrt{(E_a^L)^2 + m_b^2 + 2E_a^L m_b - |\mathbf{p}_a^L|^2}} \\ &= \frac{E_a^L + m_b}{\sqrt{m_a^2 + m_b^2 + 2E_a^l m_b}} \\ &= \frac{E_a^L + m_b}{\sqrt{(p_a + p_b)^2}} \end{aligned}$$

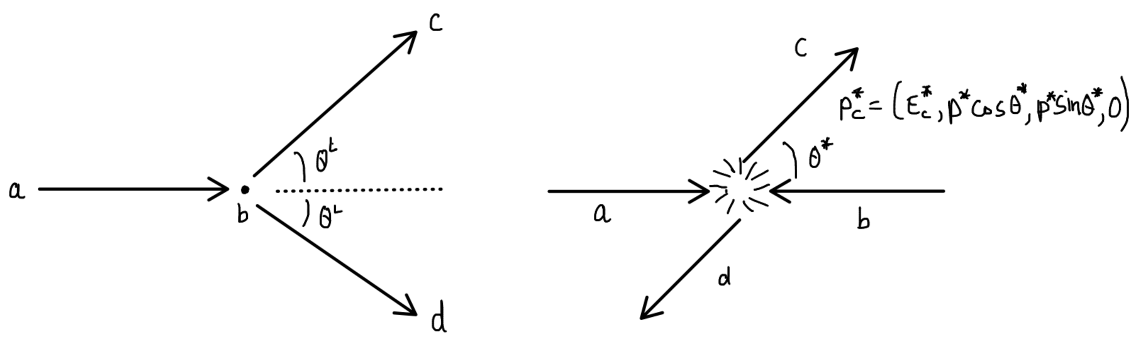
$$\boxed{\gamma_{cm} = \frac{E^L + m_b}{\sqrt{s}}} \tag{16}$$

(b) The CoM frame scattering angle θ^* and the lab-frame scattering angle θ^L are related by a boost. For elastic scattering ($m_a = m_c$, $m_b = m_d$), show that

$$\tan\theta^L = \frac{\sin\theta^*}{\gamma_{cm}(\cos\theta^* + \beta_{cm}/\beta_c^*)},$$

where β_c^* is the CoM speed of particle c.

Solution:



Lab Frame **COM Frame**

In the lab frame

$$\tan\theta^L = \frac{p_\perp^L}{p_{||}^L}$$

In the COM frame and lab frame the $p_\perp^L$ component will remain unchanged hence $p_\perp^L = p^* \sin\theta^*$ whereas

$$p_{||}^L = \gamma_{cm}(p^* \cos\theta^* + \beta_{cm} E_c^*)$$

Therefore,

$$\tan\theta^L = \frac{p^* \sin\theta^*}{\gamma_{cm}(p^* \cos\theta^* + \beta_{cm} E_c^*)}$$

$$(\text{Using } \beta_c^* = p^*/E_c^*)$$

$$\tan\theta^L = \frac{p^* \sin\theta^*}{\gamma_{cm}(p^* \cos\theta^* + \beta_{cm} p^*/\beta_c^*)}$$

$$\boxed{\tan\theta^L = \frac{\sin\theta^*}{\gamma_{cm}(\cos\theta^* + \beta_{cm}/\beta_c^*)}} \tag{17}$$

Problem 2

A photon of energy ω_1 in the lab scatters off an electron at rest (mass m_e). Let ω_2 be the energy of the scattered photon and θ the angle it makes with the beam direction.

(a) Use four-momentum conservation to derive the Compton formula:

$$\frac{1}{\omega_2} - \frac{1}{\omega_1} = \frac{1-\cos\theta}{m_e},$$

or equivalently $\lambda_2 - \lambda_1 = \frac{2\pi}{m_e}(1-\cos\theta) \equiv \lambda_C(1-\cos\theta)$, where $\lambda_C = 1/m_e$ is the Compton wavelength.

Solution:

Let p, p' denote four momentum of initial and final electron and k, k' denote four momentum of initial and final photon.

We'll do this calculation in the rest frame of the electron.

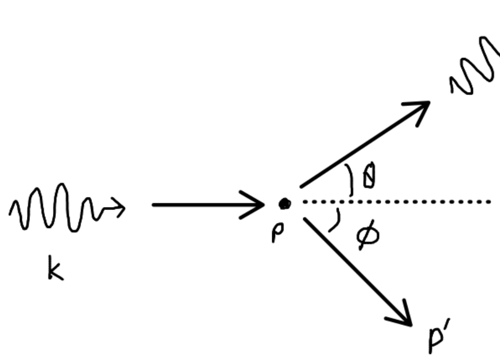


Figure 4: Compton Scattering in Lab Frame

Using conservation of four-momentum

$$\begin{aligned} p + k &= p' + k' \\ p' &= p + k - k' \end{aligned} \tag{18}$$

$$\begin{aligned} m^2 &= (p')^2 \\ &= (p + k - k')^2 \\ &= m^2 + 2p.(k - k') - 2k.k' \\ 0 &= m(\omega - \omega') - \omega\omega' + \omega\omega'\cos\theta \end{aligned}$$

$$\boxed{\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1}{m}(1-\cos\theta)} \tag{19}$$

(b) Determine the recoil electron angle ϕ in terms of θ and ω_1/m_e:

$$\cot\phi = \left(1 + \frac{\omega_1}{m_e}\right)\tan\frac{\theta}{2}.$$

Solution: We will use the invariant $p.k' = p'.k$

$$\begin{aligned} E'\omega - |\mathbf{p}'|\omega\cos\phi &= m\omega' \\ (m + (\omega - \omega'))\omega - |\mathbf{p}'|\omega\cos\phi &= m\omega' \end{aligned}$$

now use (19) and $|\mathbf{p}'|\sin\phi = \omega'\sin\theta$

$$\begin{aligned} \left(m + \frac{\omega\omega'}{m}(1-\cos\theta)\right)\omega - \omega\omega' cot\phi\sin\theta &= m\omega' \\ -\frac{\omega\omega'}{m}(1-\cos\theta)\omega + \omega\omega' cot\phi\sin\theta &= m(\omega - \omega') \\ -\frac{\omega\omega'}{m}(1-\cos\theta)\omega + \omega\omega' cot\phi\sin\theta &= m\frac{\omega\omega'}{m}(1-\cos\theta) \\ \cot\phi &= \left(1+\frac{\omega}{m}\right)\frac{(1-\cos\theta)}{\sin\theta} \end{aligned}$$

$$\boxed{\cot\phi = \left(1+\frac{\omega}{m}\right)\tan\frac{\theta}{2}} \tag{20}$$

(c) Find the kinematic limits:

- Maximum photon energy transfer (backscatter, $\theta = \pi$): show $\omega_2^{\min} = \omega_1/(1 + 2\omega_1/m_e)$.
- In the Thomson limit $\omega_1 \ll m_e$: show $\omega_2 \approx \omega_1$ (elastic scattering).
- In the extreme relativistic limit $\omega_1 \gg m_e$: show the backscattered photon energy saturates to $\omega_2^{\text{back}} \to m_e/2$.

Solution:

We have formula

$$\omega_1 - \omega_2 = \frac{\omega_1\omega_2}{m}(1-\cos\theta)$$

- for $\theta = \pi$ we have

$$\omega_1 - \omega_2^{\min} = \frac{2\omega_1\omega_2^{\min}}{m}$$
$$\omega_1 = \omega_2^{\min}\left(1 + \frac{2\omega_1}{m}\right)$$

$$\boxed{\omega_2^{\min} = \frac{\omega_1}{(1 + \frac{2\omega_1}{m})}} \tag{21}$$

- In Thomson limit $\frac{\omega_1}{m} << 1$ and hence

$$\omega_1 - \omega_2 \approx 0$$
$$\omega_1 \approx \omega_2$$

- We have

$$\omega_2 = \frac{\omega_1}{1 + \frac{\omega_1}{m}(1 - \cos\theta)}$$

for $\frac{\omega_1}{m} >> 1$ and back-scattered case the denominator

$$1 + \frac{\omega_1}{m}(1 - \cos\theta) \approx \frac{2\omega_1}{m}$$

And therefore

$$\boxed{\omega_2^{\text{back}} \to \frac{m}{2}} \tag{22}$$

1. Express the Mandelstam variables s, t, u for Compton scattering in terms of ω_1, ω_2, m_e, and θ (in the lab), and verify $s + t + u = 2m_e^2$.

   Solution:

$$s = (p + k)^2 = m^2 + 2p.k = m^2 + 2m\omega_1$$
$$t = (p - p') = -2p.p' = -2mE' = 2m^2 - 2m(m + (\omega_1 - \omega_2))$$
$$u = (p - k') = m^2 - 2p.k' = m^2 - 2m\omega'$$

   Adding all these we get

$$s + t + u = m^2 + 2m\omega_1 - 2m(\omega_1 - \omega_2) + 2m\omega_1 - 2m\omega_2$$

$$\boxed{s + t + u = 2m^2} \tag{23}$$

Problem 3

For Compton scattering $e^-(p) + \gamma(k) \to e^-(p') + \gamma(k')$:

(a) In the CoM frame, find the magnitudes of the electron and photon three-momenta as functions of s and m_e. Hence, show that:

$$E_e^* = \frac{s + m_e^2}{2\sqrt{s}}, \qquad \omega^* = \frac{s - m_e^2}{2\sqrt{s}}.$$

Solution: In COM frame
$p^\mu = (\sqrt{p^{*2} + m_e^2}, -\mathbf{p})$
$k^\mu = (|\mathbf{p}^*|, \mathbf{p}^*)$

$$\begin{aligned}
\sqrt{s} &= \sqrt{p^{*2} + m_e^2} + p^* \\
(\sqrt{s} - p^*)^2 &= p^{*2} + m_e^2 \\
s - 2p^*\sqrt{s} &= m_e^2 \\
p^* &= \frac{s - m_e^2}{2\sqrt{s}}
\end{aligned}$$

where $p^* = |\mathbf{p}|$. Therefore

$$E_e^* = p^{*2} + m_e^2$$

$$\boxed{E^{*2} = \frac{s + m_e^2}{2\sqrt{s}}} \tag{24}$$

$$\omega^* = p^*$$

$$\boxed{\omega^* = \frac{s - m_e^2}{2\sqrt{s}}} \tag{25}$$

(b) Derive the correct expressions for t and u in terms of s and θ^* (CoM scattering angle):

$$t = -\frac{(s - m_e^2)^2}{2s}(1 - \cos\theta^*), \qquad u = m_e^2 - \frac{(s - m_e^2)}{2s}\left[(s + m_e^2) + (s - m_e^2)\cos\theta^*\right].$$

Solution:

In COM there wont be any change in the frequency of the photon therefore $\omega^* = \omega'^*$

$$
\begin{aligned}
t &= (k - k')^2 \\
&= -2k.k' \\
&= -2\omega\omega' + 2\omega\omega' \cos\theta^* \\
&= -2\omega\omega'(1 - \cos\theta^*) \\
&= -2\left(\frac{s - m_e^2}{2\sqrt{s}}\right)^2 (1 - \cos\theta^*)
\end{aligned}
$$

$$
\boxed{t = -\frac{(s - m_e^2)^2}{2s}(1 - \cos\theta^*)} \tag{26}
$$

$$
\begin{aligned}
u &= (p - k')^2 \\
&= m_e^2 - 2p.k' \\
&= m_e^2 - 2E_e^*\omega^* - 2p^*\omega'^* \cos\theta^* \\
&= m_e^2 - 2\left(\frac{s + m_e^2}{2\sqrt{s}}\right)\left(\frac{s - m_e^2}{2\sqrt{s}}\right) - 2\left(\frac{s - m_e^2}{2\sqrt{s}}\right)\left(\frac{s - m_e^2}{2\sqrt{s}}\right)\cos\theta^* \\
&= m_e^2 - \left(\frac{s - m_e^2}{2s}\right)\left[(s + m_e^2) + (s - m_e^2)\cos\theta^*\right]
\end{aligned}
$$

(b) Express the lab-frame photon energy ω_1 in terms of s and m_e, and hence write the relation between θ^* (CoM) and θ (lab). Show that for $s \gg m_e^2$:

$$
\cos\theta = \frac{\cos\theta^* + \beta_{cm}}{1 + \beta_{cm}\cos\theta^*} \xrightarrow{s \gg m_e^2} 1 \quad \text{(forward collimation).}
$$

Solution:

We'll use (17) from problem 1 for process $a + b \to c + d$ where a, c is photon and b, d is electron. Since c is photon $\beta_c = 1$

$$
\begin{aligned}
\tan\theta &= \frac{\sin\theta^*\sqrt{1 - \beta_{cm}^2}}{\cos\theta^* + \beta_{cm}} \\
\frac{1 - \cos^2\theta}{\cos^2\theta} &= \frac{\sin^2\theta^*(1 - \beta_{cm}^2)}{(\cos\theta^* + \beta_{cm})^2} \\
\cos^2\theta &= \frac{(\cos\theta^* + \beta_{cm})^2}{\cos^2\theta^* + \beta^2 + 2\cos\theta^*\beta + \sin^2\theta^* - \beta^2\sin^2\theta^*}
\end{aligned}
$$

$$\boxed{\cos\theta = \frac{\cos\theta^* + \beta_{cm}}{1 + \beta_{cm}\cos\theta^*}} \tag{27}$$

for $s >> m_e$ particle will be ultra-relativistic and $\mathbf{p}_\gamma^L = E_\gamma$ and $E_\gamma >> m_e$ therefore from (15) $\beta_{cm} \sim 1$
, which gives

$$\boxed{\cos\theta \to 1}$$

(c) What is the maximum recoil energy imparted to the electron in the lab frame? Express it in terms of ω_1 and m_e, and show that this defines the Compton edge at

$$T_e^{\max} = \frac{2\omega_1^2}{m_e + 2\omega_1}.$$

Solution: Using Energy Conservation

$$\begin{aligned} E + \omega_1 &= E' + \omega_2 \\ m + \omega_1 &= T + m + \omega_2 \\ T_{\max} &= (\omega_1 - \omega_2)\Big|_{\max} \end{aligned}$$

Using (21) we get

$$\begin{aligned} (\omega_1 - \omega_2)\Big|_{\max} &= \omega_1 - \frac{\omega_1}{1 + \frac{2\omega_1}{m}} \\ &= \frac{m\omega_1 + 2\omega_1^2 - m\omega_1}{m + 2\omega_1} \end{aligned}$$

$$\boxed{T_{\max} = \frac{2\omega_1^2}{m_e + 2\omega_1}} \tag{28}$$

Problem 4

Consider unpolarised Compton scattering $e^-(p,s)\,\gamma(k,\lambda) \to e^-(p',s')\,\gamma(k',\lambda')$. The two tree-level diagrams give

$$\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u,$$

$$\mathcal{M}_s = -e^2\,\bar{u}(p')\,\not{\epsilon}'^*\,\frac{\not{p} + \not{k} + m}{s - m^2}\,\not{\epsilon}\,u(p), \qquad \mathcal{M}_u = -e^2\,\bar{u}(p')\,\not{\epsilon}\,\frac{\not{p} - \not{k}' + m}{u - m^2}\,\not{\epsilon}'^*\,u(p).$$

(a) Using the Dirac equation $\bar{u}(p')(\not{p}'-m)=0$ and the gauge condition $k\cdot\epsilon = k'\cdot\epsilon' = 0$, simplify the numerators of $\mathcal{M}_s$ and $\mathcal{M}_u$ using the identity

$$\not{k}\not{\epsilon} = 2k\cdot\epsilon - \not{\epsilon}\not{k}.$$

Solution:

$$\begin{aligned}
\mathcal{M}_s &= -e^2\bar{u}(p')\epsilon^*_\mu\epsilon_\nu \frac{\gamma^\mu\not{p}\gamma^\nu + \gamma^\mu\not{k}\gamma^\nu + \gamma^\mu m\gamma^\nu}{(p+k)^2-m^2}u(p) \\
&= -e^2\bar{u}(p')\epsilon^*_\mu\epsilon_\nu \frac{\gamma^\mu(2p^\nu - \gamma^\nu\not{p}) + \gamma^\mu\not{k}\gamma^\nu + \gamma^\mu m\gamma^\nu}{2p.k}u(p) \\
&= -e^2\bar{u}(p')\epsilon^*_\mu\epsilon_\nu \frac{2\gamma^\mu p^\nu + \gamma^\mu\not{k}\gamma^\nu}{2p.k}u(p)
\end{aligned}$$

Similarly in case of $\mathcal{M}_u$ we will have denominator $(p-k')^2 = -2p.k'$ and μ and ν interchanged in gammas

$$\mathcal{M}_u = -e^2\bar{u}(p')\epsilon^*_\mu\epsilon_\nu \frac{-\gamma^\nu\not{k}'\gamma^\mu + 2\gamma^\nu p^\mu}{-2p.k'}$$

We get total amplitude

$$\boxed{\mathcal{M} = -e^2\epsilon^*_\mu\epsilon_\nu\bar{u}(p')\left[\frac{2\gamma^\mu p^\nu + \gamma^\mu\not{k}\gamma^\nu}{2p.k} - \frac{2\gamma^\nu p^\mu - \gamma^\nu\not{k}'\gamma^\mu}{2p.k'}\right]u(p)} \tag{29}$$

(b) Verify that $\mathcal{M}$ is gauge invariant: replacing $\epsilon^\mu \to \epsilon^\mu + \lambda k^\mu$ leaves $\mathcal{M}$ unchanged. Show this requires using the Ward identity and Dirac equation. Is each diagram individually gauge invariant?

Solution: First lets just see change in $\mathcal{M}_s$

$$\begin{aligned}
\mathcal{M}_s \to \mathcal{M}'_s &= -e^2(\epsilon^*_\mu + \lambda k^*_\mu)(\epsilon_\nu + \lambda k_\nu)\bar{u}(p')\left[\frac{2\gamma^\mu p^\nu + \gamma^\mu\not{k}\gamma^\nu}{2p.k} - \frac{2\gamma^\nu p^\mu - \gamma^\nu\not{k}'\gamma^\mu}{2p.k'}\right]u(p) \\
&= \mathcal{M}_s - e^2(\lambda\epsilon^*_\mu k_\nu + \lambda\epsilon_\nu k^*_\mu + \lambda^2 k^*_\mu k_\nu)\left[\frac{2\gamma^\mu p^\nu + \gamma^\mu\not{k}\gamma^\nu}{2p.k} - \frac{2\gamma^\nu p^\mu - \gamma^\nu\not{k}'\gamma^\mu}{2p.k'}\right]u(p)
\end{aligned}$$

we'll use $\not{k}\not{k} = k^2 = 0$ hence we get

$$\mathcal{M}'_s = \mathcal{M}_s - e^2$$

(c) Show that the spin-and-polarisation-averaged amplitude squared is

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{2}\left[\frac{M_{ss}}{(s-m^2)^2} + \frac{M_{uu}}{(u-m^2)^2} + \frac{M_{su}+M_{us}}{(s-m^2)(u-m^2)}\right],$$

and identify each piece as a squared diagram plus interference term.

Solution:

(d) Evaluate $\overline{|\mathcal{M}|^2}$ by computing the necessary traces. Show that in the massless limit ($m_e = 0$) the result reduces to

$$\overline{|\mathcal{M}|^2}\Big|_{m_e=0} = -2e^4\left(\frac{s}{u} + \frac{u}{s}\right).$$

Check that this is positive for physical Compton kinematics ($s > 0$, $u < 0$).

Solution:

Problem 5

1. Using the general two-body differential cross-section formula and the Compton kinematics in the lab frame, show that

$$\frac{d\sigma}{d\Omega_L} = \frac{\alpha^2}{2m_e^2}\left(\frac{\omega_2}{\omega_1}\right)^2\left[\frac{\omega_1}{\omega_2} + \frac{\omega_2}{\omega_1} - \sin^2\theta\right].$$

This is the **Klein–Nishina formula** (1929).

Solution:

2. In the low-energy (Thomson) limit $\omega_1 \ll m_e$, show this reduces to

$$\frac{d\sigma}{d\Omega}\Big|_{\text{Thomson}} = r_e^2\left(1 - \sin^2\theta\cos^2\phi\right),$$

where $r_e = \alpha/m_e$ is the classical electron radius. For unpolarised photons, show the angular distribution becomes

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2}(1 + \cos^2\theta).$$

Solution:

3. Integrate the unpolarised Thomson formula to obtain the total Thomson cross-section:

$$\sigma_{\text{Thomson}} = \frac{8\pi}{3} r_e^2 \approx 0.665 \text{ barn}.$$

Solution:

4. Show that at high energies $\omega_1 \gg m_e$, the total Klein–Nishina cross-section falls as

$$\sigma_{KN} \xrightarrow{\omega_1 \gg m_e} \frac{\pi\alpha^2}{m_e\omega_1} \ln \frac{2\omega_1}{m_e}.$$

Comment on the physical significance of this $1/\omega_1$ behaviour.

Solution:

Problem 6

In inverse Compton scattering (ICS), a low-energy photon scatters off a relativistic electron ($\gamma_e = E_e/m_e \gg 1$). Let the photon have energy ϵ in the lab and the electron have Lorentz factor γ_e.

1. Show that in the electron rest frame, the photon energy is boosted to $\epsilon^* \approx \gamma_e\epsilon(1 + \beta\cos\alpha)$, where α is the angle of incidence in the lab. For a head-on collision:

$$\epsilon^* \approx 2\gamma_e\epsilon.$$

Solution:

2. In the Thomson regime of ICS ($\epsilon^* \ll m_e$), show that the maximum upscattered photon energy in the lab is:

$$\epsilon'_{\max} \approx 4\gamma_e^2\,\epsilon.$$

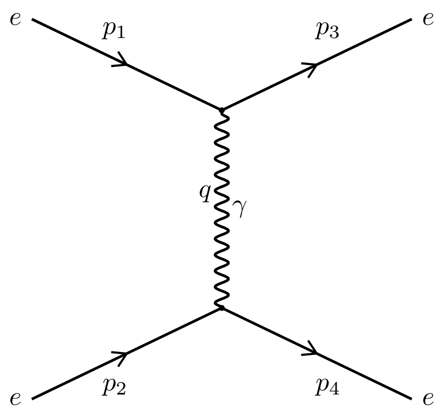
Estimate $\epsilon'_{\max}$ for a 10 GeV electron scattering a CMB photon ($\epsilon \sim 6 \times 10^{-4}$ eV).

Solution:

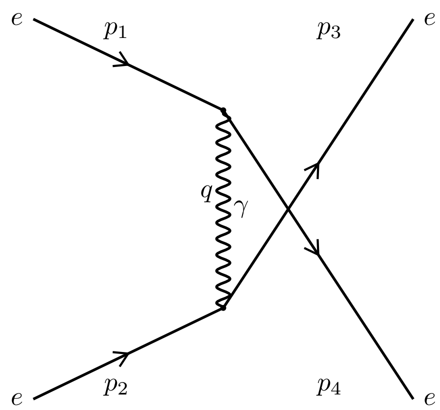
Problem 7

(a) Draw all contributing tree-level Feynman diagrams for $e^-(p_1)\, e^-(p_2) \to e^-(p_3)\, e^-(p_4)$ and write down the full invariant amplitude, paying careful attention to the relative sign from Fermi statistics.

Solution:



t-channel, $\quad q = p_1 - p_3$



u-channel, $\quad q = p_1 - p_4$

for the t-channel process the contractions will be,

$$\langle e_1^- e_2^- | \bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\bar{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|e_3^- e_4^-\rangle \tag{30}$$

here the sign will be $(-1)^3 = -1$ to untangle all contractions
for the u-channel process the contraction will be

$$\langle e_1^- e_2^- | \bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\bar{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|e_3^- e_4^-\rangle \tag{31}$$

here the sign will be $(-1)^2 = 1$
Hence there will be a relative minus sign between the two diagrams
The Amplitude now can be written as

$$\begin{aligned} i\mathcal{M} &= \bar{u}(p_3)(-ie\gamma^\mu)u(p_1)\left(\frac{-ig_{\mu\nu}}{(p_1-p_3)^2}\right)\bar{u}(p_4)(-ie\gamma^\nu)u(p_2) \\ &\quad - \bar{u}(p_3)(-ie\gamma^\nu)u(p_2)\left(\frac{-ig_{\mu\nu}}{(p_1-p_4)^2}\right)\bar{u}(p_4)(-ie\gamma^\mu)u(p_1) \end{aligned}$$

$$\boxed{i\mathcal{M} = ie^2\left[\frac{\bar{u}(p_3)\gamma^\mu u(p_1)\bar{u}(p_4)\gamma_\mu u(p_2)}{(p_1-p_3)^2} - \frac{\bar{u}(p_3)\gamma^\mu u(p_2)\bar{u}(p_4)\gamma_\mu u(p_1)}{(p_1-p_4)^2}\right]} \tag{32}$$

(b) Show that in the massless-electron limit:

$$\overline{|\mathcal{M}|^2} = 2e^4 \left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{2s^2}{tu}\right].$$

Explain the origin of the $2s^2/tu$ interference term and its sign.

Solution:

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{4}\sum_{\text{spins}}\left[\frac{\bar{u}(p_3)\gamma^\mu u(p_1)\bar{u}(p_4)\gamma_\mu u(p_2)}{(p_1-p_3)^2} - \frac{\bar{u}(p_3)\gamma^\mu u(p_2)\bar{u}(p_4)\gamma_\mu u(p_1)}{(p_1-p_4)^2}\right]\times$$
$$\left[\frac{\bar{u}(p_2)\gamma^\nu u(p_4)\bar{u}(p_1)\gamma_\nu u(p_3)}{(p_1-p_3)^2} - \frac{\bar{u}(p_1)\gamma^\nu u(p_4)\bar{u}(p_2)\gamma_\nu u(p_3)}{(p_1-p_4)^2}\right]$$

We'll evaluating this term by term we get

$$\begin{aligned}\mathbf{I} &= \sum_{\text{spins}} \bar{u}(p_3)_a(\gamma^\mu)_{ab}u(p_1)_b\bar{u}(p_4)_c(\gamma_\mu)_{cd}u(p_2)_d\bar{u}(p_2)_e(\gamma^\nu)_{ef}u(p_4)_f\bar{u}(p_1)_g(\gamma_\nu)_{gh}u(p_3)_h \\ &= \text{tr}\left[(\not{p}_3+m)\gamma^\mu(\not{p}_1+m)\gamma^\nu\right]\text{tr}\left[(\not{p}_4+m)\gamma_\mu(\not{p}_2+m)\gamma_\nu\right] \\ &= 32\left[(p_1.p_2)(p_3.p_4) - m^2(p_2.p_4) + (p_2.p_3)(p_1.p_4) - m^2(p_1.p_3) + 2m^4\right]\end{aligned}$$

using invariants $p_1.p_3 = p_2.p_4$, $p_1.p_2 = p_3.p_4$ and $p_1.p_4 = p_2.p_3$ we get final result as

$$\mathbf{I} = \frac{32[(p_1.p_2)^2 + (p_2.p_3)^2 - 2m^2(m^2 - p_1.p_3)]}{(p_1-p_3)^4} \tag{33}$$

Similarly we'll evaluate second and third trace which are identical

$$\begin{aligned}\mathbf{II} &= \text{tr}\left[(\not{p}_3+m)\gamma^\mu(\not{p}_1+m)\gamma^\nu(\not{p}_4+m)\gamma_\nu(\not{p}_2+m)\gamma_\nu\right] \\ &= 32\left[-(p_1.p_2)(p_3.p_4) + m^2p_2.p_4 + m^2p_1.p_2 + m^2p_1.p_4 - m^4\right]\end{aligned}$$

Evaluating this using invariants we get

$$\boxed{\mathbf{II} = \mathbf{III} = \frac{32(p_1.p_2)[2m^2 - p_1.p_2]}{(p_3-p_1)^2(p_4-p_1)^2}} \tag{34}$$

Looking at the **IV** we have to just interchange $p_1 \leftrightarrow p_2$

$$\mathbf{IV} = \frac{32[(p_1.p_2)^2 + (p_1.p_3)^2 - 2m^2(m^2 - p_2.p_3)]}{(p_1-p_4)^4} \tag{35}$$

Adding all these traces from (33), (34) and (35) and using $p_1.p_2 = \frac{s}{2} - m^2$, $p_1.p_3 =$

$m^2 - \frac{t}{2}$ and $p_1.p_3 = m^2 - \frac{u}{2}$

$$\overline{|\mathcal{M}|^2} = 8e^4 \Bigg\{ \frac{\left(\frac{s}{2} - m^2\right)^2 + \left(\frac{t}{2} - m^2\right)^2 - 2m^2\left(m^2 - \frac{u}{2} + m^2\right)}{4\left(\frac{t}{2} - m^2\right)^2} - \frac{2\left(\frac{s}{2} - m^2\right)\left(2m^2 - \frac{s}{2} + m^2\right)}{4\left(\frac{t}{2} - m^2\right)\left(\frac{u}{2} - m^2\right)}$$
$$+ \frac{\left(\frac{s}{2} - m^2\right)^2 + \left(\frac{u}{2} - m^2\right)^2 - 2m^2\left(m^2 - \frac{t}{2} + m^2\right)}{4\left(\frac{u}{2} - m^2\right)^2} \Bigg\} \tag{36}$$

In the massless limit this becomes

$$\boxed{\overline{|\mathcal{M}|^2} = 2e^4 \left\{ \frac{s^2 + t^2}{t^2} + \frac{2s^2}{ut} + \frac{s^2 + u^2}{u^2} \right\}} \tag{37}$$

The interference term has positive sign as both diagrams describe the same repulsive process which enhances the interference

(c) In the CoM frame for massless electrons, let θ be the scattering angle. Express s, t, u in terms of the CoM energy $E = \sqrt{s}/2$ and θ. Hence, find $d\sigma/d\Omega$ and explain the forward/backward divergence.

Solution:

$$s = 4E^2$$

$$\begin{aligned} t &= (p_1 - p_3)^2 \\ &= (0, E - E\cos\theta, -E\sin\theta)^2 \\ &= -2E^2(1 - \cos\theta) \end{aligned}$$

$$u = (p_1 - p_4)^2 = -2E^2(1 + \cos\theta)$$

$$\begin{aligned} \overline{|\mathcal{M}|^2} &= 2e^4 \left\{ \frac{16 + 4(1 - \cos\theta)^2}{4(1 - \cos\theta)^2} + \frac{2 \times 16}{4\sin^2\theta} + \frac{16 + 4(1 + \cos\theta)^2}{4(1 + \cos\theta)^2} \right\} \\ &= 4e^4 \left\{ 1 + \frac{4(1 + \cos^2\theta)}{\sin^4\theta} + \frac{8}{\sin^2\theta} \right\} \end{aligned}$$

Since all four particles are identical the differential cross section will be

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{CM} = \frac{1}{2} \frac{\overline{|\mathcal{M}|^2}}{64\pi^2 E_{\mathrm{CM}}^2}$$

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{CM} = \frac{\alpha^2}{8E^2} \left\{ 1 + \frac{4(1 + \cos^2\theta)}{\sin^4\theta} + \frac{8}{\sin^2\theta} \right\} \tag{38}$$

When we put $\theta = 0$ or $\theta = \pi$ there is divergence in the differential cross section

- When $\theta = 0$
  The photon exchanged in the t-channel carries zero momentum, the electrons barely deflect each other at large distances. This is the infrared divergence of the long-range Coulomb force. The $1/t^2 \sim 1/\sin^4\theta$ *is exactly the Rutherford divergence.*

- *when* $\theta = \pi$
  Divergence is the direct signature of the electrons being identical — it is the u-channel Coulomb pole entering the physical region purely because you cannot label which electron is which.

(d) Show that in the non-relativistic limit, Møller scattering reduces to

$$\frac{d\sigma}{d\Omega} \to \frac{\alpha^2}{4m_e^2 v^4 \sin^4(\theta/2)},$$

and explain why the result at $\theta = \pi/2$ is twice the Rutherford cross-section.

Solution:

For non-relativistic case: $E \approx m$ and therefore,

$$s \approx 4m^2$$

$$\begin{aligned} t &= (p_1 - p_3)^2 \\ &\approx -2m^2v^2(1 - \cos\theta) \end{aligned}$$

$$u \approx -2m^2v^2(1 + \cos\theta)$$

also $s >> |t|, |u|$ we get

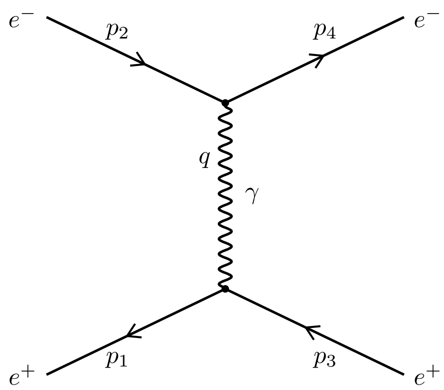
$$\boxed{\overline{|\mathcal{M}|^2} \approx 2e^4 \left\{ \frac{s^2}{t^2} + \frac{2s^2}{ut} + \frac{s^2}{u^2} \right\}}$$

$$
\begin{aligned}
\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{CM} &= \frac{\alpha^2}{8m^2}\left\{\frac{1}{4m^4v^4(1-\cos\theta)^2}+\frac{2}{4m^4v^4(1-\cos^2\theta)+}+\frac{1}{4m^4v^4(1+\cos\theta)^2}\right\}\\
&= \frac{\alpha^2}{4m^2v^4}\frac{1}{\sin^4\frac{\theta}{2}\cos^4\frac{\theta}{2}}\\
&\text{(for small angles we have)}\\
&= \frac{\alpha^2}{4m^2v^4\sin^4\frac{\theta}{2}}
\end{aligned}
$$

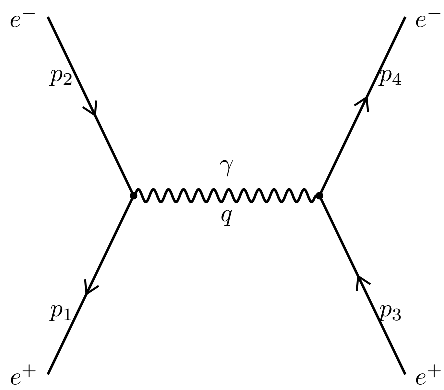
Problem 8

(a) Draw all tree-level Feynman diagrams for $e^+(p_1)\,e^-(p_2)\to e^+(p_3)\,e^-(p_4)$ and write down $\mathcal{M}$. Explain why the relative sign between s- and t-channel diagrams differs from the Møller case.

Solution:



t-channel, $\quad q = p_1 - p_3$



s-channel, $\quad q = p_1 + p_2$

for s channel process we'll have contractions

$$
\langle e_1^+ e_2^- |\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\overline{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|e_3^+ e_4^-\rangle \tag{39}
$$

with sign of $(-1)^1 = -1$ another contractions give t-channel

$$
\langle e_1^+ e_2^- |\bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\overline{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|e_3^+ e_4^-\rangle \tag{40}
$$

with relative sign of $(-1)^4 = 1$

$$\begin{aligned}
i\mathcal{M} =& \bar{v}(p_1)(-ie\gamma^\mu)u(p_2)\left(\frac{-ig_{\mu\nu}}{(p_1+p_2)^2}\right)\bar{u}(p_4)(-ie\gamma^\nu)v(p_3) \\
&- \bar{u}(p_4)(-ie\gamma^\mu)u(p_2)\left(\frac{-ig_{\mu\nu}}{(p_1-p_3)^2}\right)\bar{v}(p_1)(-ie\gamma^\nu)v(p_3)
\end{aligned}$$

$$\boxed{i\mathcal{M} = ie^2\left\{\frac{\bar{v}(p_1)\gamma^\mu u(p_2)\bar{u}(p_4)\gamma_\mu v(p_3)}{(p_1+p_2)^2} - \frac{\bar{u}(p_4)\gamma^\mu u(p_2)\bar{v}(p_1)\gamma_\mu v(p_3)}{(p_1-p_3)^2}\right\}} \tag{41}$$

The s-channel diagram requires swapping of a fermion line relative to the t-channel hence a minus sign.

(b) Show that in the massless limit:

$$\overline{|\mathcal{M}|^2} = 2e^4\left[\frac{t^2+u^2}{s^2} + \frac{s^2+u^2}{t^2} - \frac{2u^2}{st}\right].$$

Compare with the Moller result and explain the sign of the interference term.

Solution:

To derive this we'll use the previous Moller scattering result (37) and crossing symmetry, which tells us to interchange $p_3 \leftrightarrow -p_1$ therefore earlier $s = (p_1+p_2)^2 \to (p_2-p_3)^2 = u$, $u = (p_1-p_4)^2 \to (p_3+p_4)^2 = s$ and $t = (p_1-p_3)^2 \to -(p_1-p_3)^2 = -t$. Therefore,

$$\boxed{\overline{|\mathcal{M}|}^2 = 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{u^2+t^2}{s^2} - \frac{2u^2}{ts}\right]} \tag{42}$$

(c) Show that the ratio of total Bhabha to Moller cross-sections approaches unity in the high-energy limit, and explain why in terms of helicity conservation.

Solution:

(d) At low energies ($\sqrt{s} \ll m_e$), show that both Bhabha and Møller cross-sections approach the Mott cross-section:

$$\frac{d\sigma_{\text{Mott}}}{d\Omega} = \frac{\alpha^2}{4m_e^2v^4\sin^4(\theta/2)}\cos^2(\theta/2).$$

Solution:

Problem 9

Consider $e^+(p_1)\,e^-(p_2) \to \gamma(k_1)\,\gamma(k_2)$ in the lab where the electron is at rest.

(a) Draw the two Feynman diagrams and write the amplitude.

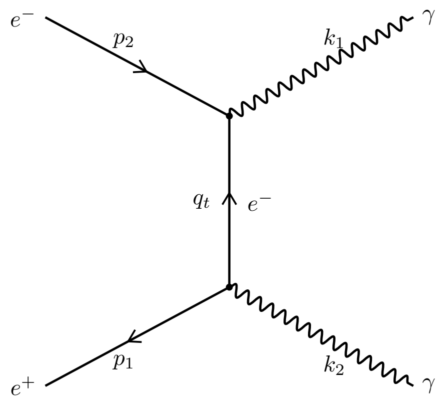
Solution:

First we'll figure out the possible contractions

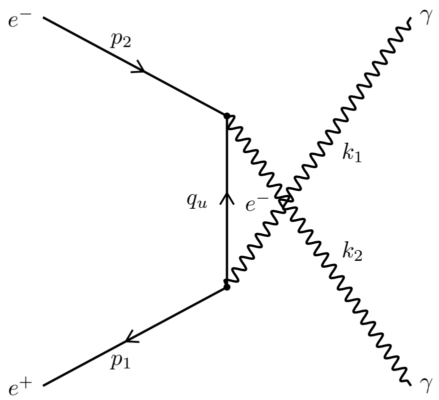
$$\langle e_1^+ e_2^- | \bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\overline{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|\gamma_1\gamma_2\rangle$$

$$\langle e_1^+ e_2^- | \bar{\psi}(x)\gamma^\mu A_\mu(x)\psi(x)\overline{\psi}(y)\gamma^\mu A_\mu(y)\psi(y)|\gamma_1\gamma_2\rangle$$

These gives t-channel and u-channel process



t-channel, $\quad q_t = p_2 - k_1$



u-channel, $\quad q_u = p_2 - k_2$

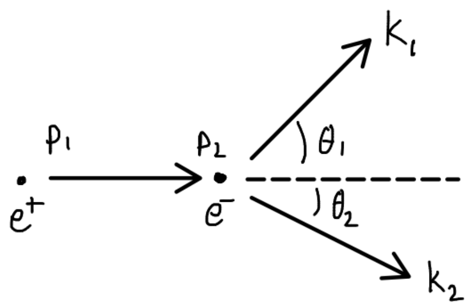
$$\begin{aligned}
i\mathcal{M} &= \bar{v}(p_1)(-ie\gamma^\mu)\frac{i(\not{p}_2 - \not{k}_1 + m)}{(p_2-k_1)^2 - m^2}(-ie\gamma^\nu)u(p_2)\epsilon^*_\mu(k_2)\epsilon^*_\nu(k_1) \\
&\quad + \bar{v}(p_1)(-ie\gamma^\mu)\frac{i(\not{p}_2 - \not{k}_2 + m)}{(p_2-k_2)^2 - m^2}(-ie\gamma^\nu)u(p_2)\epsilon^*_\mu(k_1)\epsilon^*_\nu(k_2) \\
&= -e^2\epsilon^*_\mu(k_1)\epsilon^*_\nu(k_2)\bar{v}(p_1)\left[\gamma^\mu\frac{i(\not{p}_2 - \not{k}_2 + m)}{-2p_2.k_2}\gamma^\nu + \gamma^\nu\frac{i(\not{p}_2 - \not{k}_1 + m)}{-2p_2.k_1}\gamma^\mu\right]u(p_2)
\end{aligned}$$

(b) In the lab frame (electron at rest, positron energy E_+, momentum p_+), derive:

$$\omega_1 = \frac{m_e(E_+ + m_e)}{E_+ + m_e - p_+\cos\theta_1}, \qquad \omega_2 = \frac{m_e(E_+ + m_e)}{E_+ + m_e - p_+\cos\theta_2}.$$

What is the angular constraint between θ_1, θ_2, and the transverse momenta?

Solution:



$$p_1 = (m, 0, 0, 0)$$

$$p_2 = (E_+, p_+, 0, 0)$$

$$k_1 = (\omega_1, \omega_1 \cos\theta_1, \omega_1 \sin\theta_1, 0)$$

$$k_2 = (\omega_2, -\omega_2 \cos\theta_2, -\omega_2 \sin\theta_2, 0)$$

by conservation of 4-momentum

$$\begin{aligned} p_1 + p_2 &= k_1 + k_2 \\ p_1 + p_2 - k_2 &= k_1 \\ &\text{Square now} \\ m^2 + p_1.p_2 - k_1.p_1 - k_1.p_2 &= 0 \\ m^2 + mE_+ - m\omega_1 - E_+\omega_1 + p_+\omega_1 \cos\theta_1 &= 0 \end{aligned}$$

$$\boxed{\omega_1 = \frac{m_e(m_e + E_+)}{m_2 + E_+ - p_+ \cos\theta_1}} \tag{43}$$

similarly going other way around

$$p_1 + p_2 - k_2 = k_1$$

we'll get

$$\boxed{\omega_2 = \frac{m_e(m_e + E_+)}{m_2 + E_+ - p_+ \cos\theta_2}} \tag{44}$$

(c) In the at-rest limit ($E_+ = m_e$, $p_+ = 0$), show both photons are emitted back-to-back with $\omega_1 = \omega_2 = m_e = 0.511$ MeV. Explain the significance for positron emission tomography (PET).

Solution:
Using (43) and (44) and putting $p_+ = 0$ we'll get $\boldsymbol{\omega_1 = \omega_2 = m_e}$
Positron Emission Tomography:

- A PET scan is an imaging test that uses a small amount of radioactive material, known as a radiotracer, to evaluate the metabolic activity of cells in the body. This technique allows healthcare providers to observe how organs and tissues are functioning in real time in 3-dimensions.
- Tracers are specially designed radioactive molecules made using cyclotron and biological molecule synthesizer
- Tracer is injected in patients bloodstream and the organ to be scanned absorbs the tracer
- Inside organ the radioactive atom in tracer gives of an positron which annihilates the electron nearby releasing 2 gamma rays in opposite direction
- The pet scanner when detects gammas on two opposite side can detect from which part of the body gamma came
- By detecting many of these processes and some processing the organs activity can be detected in 3-dimensions
- Different tracers can detect **Cancer**, test **heart function**, track **Alzheimer**

(d) Show that the unpolarised annihilation cross-section from rest (in the NR limit, $v \to 0$) is:

$$\sigma(e^+e^- \to \gamma\gamma) \xrightarrow{v\to 0} \frac{\pi r_e^2}{v},$$

and comment on the $1/v$ threshold behaviour.

Solution: